PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Determinants of Drug Prices: A Systematic Review of Comparison Studies
AUTHORS	Janssen Daalen, Jules; den Ambtman, Anouk; Van Houdenhoven, Mark; van den Bemt, Bart

VERSION 1 – REVIEW

REVIEWER	Lakdawalla, Darius University of Southern California
REVIEW RETURNED	19-Dec-2020

GENERAL COMMENTS	The authors conduct a systematic review of the literature to determine the effect of specific policies on net pharmaceutical prices across countries. The study is competently executed, but significance is limited for several reasons: 1. As the authors note, the literature on net prices is virtually absent. This is a well-known feature of the academics research on the pharmaceutical industry. However, since payers care about net prices rather than retail prices, this limits the significance of the study. 2. Prior research suggests that the effects of individual policies – on retail prices – varies depending on the other policies that have been passed in the country. (See Sood N, et al. The effect of regulation on pharmaceutical revenues: experience in nineteen countries. Health Affairs 2009;28; w125-137.) In particular, this research finds that the first policy implemented has substantially larger effects on subsequent policies, and the order of implementation matters more than the precise nature of the policy. Thus, the significance of studying the effects of specific policies seems limited. 3. There is already quite a large literature in health economics on the determinants of drug prices – notably, however, it too consists primarily of studies on gross, rather than net prices. (For one review of the literature, see Lakdawalla DN. Economics of the pharmaceutical industry. Journal of Economic Literature 2018;56; 397-449.) 4. From the economics literature, it is well-understood that the negotiating leverage of the payer reduces prices. In other words, the ability of the payer to credibly walk away from the deal, or force the launch of the drug at a lower price, substantially influences the price negotiation. This implies, however, that the same policy will have different effects when implemented in markets of different size and negotiating leverage. This presents another reason why a

	given policy might have heterogenous effects - this time across countries rather than by order of implementation. And, this in turn limits the significance of a review that is designed to assess the average effectiveness of particular policies.

REVIEWER	Mattingly II, T. Joseph University of Maryland School of Pharmacy, Pharmacy Practice and Science
REVIEW RETURNED	21-Dec-2020

GENERAL COMMENTS	First, thank you for submitting this review to the journal. Consider the following to help advance your contribution through the peer review process: 1. Title/Objective - This article is described as an "international benchmark systematic review" which is very confusing since the methods include articles that merely compare drug prices. A more accurate reflection may simply be a review of drug price comparison studies. Especially since you included the "national" studies that don't really compare across countries (e.g. How does a phone survey of 13 pharmacies by Scott et al. asking for the "cash"/retail price meet your inclusion criteria?). 2. Grey Literature - Peer-reviewed academic journal articles may not be the most appropriate source of information regarding drug pricing comparisons. Government, non-profit organization, and other organizational reports may actually be more valuable than evidence aimed at academics. 3. Retail vs. Out-of-pocket Price - Scott et al. 2019 and Theisen et al. 2019 literally did the same thing: "called pharmacies to ask for price for a cash-paying patient". However, you categorize them differently? Why? 4. Study Quality - Scott et al. 2019 made phone calls to pharmacies across 5 US cities (Boston, Charlotte, Denver, Detroit, and Seattle) but made no price adjustments based on purchasing power. Based on your stated quality rating system, it seemed a bit surprising this meet all quality criteria to earn a "good" rating (same for Theisen et al). 5. Conclusions - "Uniform introduction of pricing regulation and publication of net drug prices will lead to more price transparency..." - How do you conclude this based on what you did? You reviewed peer-reviewed papers of other people who compared drug prices. While the statement may be true, I'm confused how this statement made it into the conclusion of your systematic review. It sounds like something that is more appropriate in your discussion.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Prof. Darius Lakdawalla, University of Southern California

Comments to the Author:

The authors conduct a systematic review of the literature to determine the effect of specific policies on net pharmaceutical prices across countries. The study is competently executed, but significance is limited for several reasons.

Response: we thank professor Lakdawalla for his thoughtful comments which have improved our manuscript. We realized that our wordings could give the suggestion that we compare prices (in that case net prices were essential) However, this study aimed to assess determinants for price differences.

1. As the authors note, the literature on net prices is virtually absent. This is a well-known feature of the academics research on the pharmaceutical industry. However, since payers care about net prices rather than retail prices, this limits the significance of the study.

Response:
First and foremost, we realized that we could be clearer that this study aims to give an overview of the current state-of-the-literature on determinants of drug prices, not the prices itself.
One of our conclusions is that availability of observational studies on determinants is still highly limited, and we recognize the limitation of near-complete absence of data on net drug prices in our results and discussion. However, the drug market is ever-expanding and the problem of rising medication expenditure is more contemporary than ever due to market entry of developing countries and new innovative therapies. Therefore, there is a desperate need for academic attention regarding the determinants, specifically modifiable determinants, of drug prices in a national and international context. Although indeed there is virtually no data on determinants of net drug prices available, we believe that identification of major determinants is possible with mostly retail or list prices.
We have now clarified throughout the entire manuscript the aim of this study.

Manuscript changes, for example:

Abstract:
“Objective: In this systematic review on drug price comparison studies, we report on recent determinants of drug prices in a national and international context to facilitate the regulation of drug prices by purchasers and policymakers worldwide. Determinants of drug prices were divided into non-modifiable and modifiable and categorized as pertaining to a country’s income level, pharmaceutical market system and its policies and government.
Primary outcome: determinants of drug prices or price variance.”

Introduction:
“Therefore, in this systematic review, we aim to identify determinants of drug prices and more specifically modifiable determinants that may serve as potential targets for drug price optimization. In addition, we give an overview of the recent literature in drug price comparisons and execute a best-evidence synthesis of all reported determinants of drug prices.”

3. There is already quite a large literature in health economics on the determinants of drug prices – notably, however, it too consists primarily of studies on gross, rather than net prices. (For one review of the literature, see Lakdawalla DN. Economics of the pharmaceutical industry. Journal of Economic Literature 2018;56; 397-449.)

Response: we are aware of this extensive and excellent multidimensional paper, which on various domains covers subjects of the current manuscript such as the effects of regulation, market size and market entry. To this extent, it provides a background to our manuscript, which additionally provides a

comprehensive overview of the current evidence of determinants and corresponding measures of drug prices. We now incorporate this review in our discussion, as part of a section on the dynamics of interventions, to put our manuscript in a broader context.

Manuscript changes (Discussion): “An earlier extensive review on determinants of pharmaceutical expenditure has also shed light on these dynamic properties of market interventions and governmental regulation on drug prices (as well as on investments in research and development). [44]”

We have combined the response to questions 2 and 4.

2. Prior research suggests that the effects of individual policies – on retail prices – varies depending on the other policies that have been passed in the country. (See Sood N, et al. The effect of regulation on pharmaceutical revenues: experience in nineteen countries. Health Affairs 2009;28; w125-137.) In particular, this research finds that the first policy implemented has substantially larger effects on subsequent policies, and the order of implementation matters more than the precise nature of the policy. Thus, the significance of studying the effects of specific policies seems limited.

AND

4. From the economics literature, it is well-understood that the negotiating leverage of the payer reduces prices. In other words, the ability of the payer to credibly walk away from the deal, or force the launch of the drug at a lower price, substantially influences the price negotiation. This implies, however, that the same policy will have different effects when implemented in markets of different size and negotiating leverage. This presents another reason why a given policy might have heterogenous effects - this time across countries rather than by order of implementation. And, this in turn limits the significance of a review that is designed to assess the average effectiveness of particular policies.

Response: we would like to stress that our review is not meant to quantify the effectiveness of policies, as our study is not designed for this aim, but to summarize and overview the peer-reviewed evidence for all measures/determinants that have been published in the last fifteen years.
In our discussion however, we do recognize that the context of the policy measure is of paramount importance and discuss the variability of outcome in different contexts of generics and originator market portions and tendering (in the paragraph ‘Turning modifiable determinants into policies’).
We have now expanded this in our discussion section. In addition, we have added a new essential component to our future research recommendations, activated by the reviewer’s comments: careful assessment of external validity of results.

Manuscript changes: “All in all, no intervention should be evaluated on its own; new interventions need to be weighted in the total ecosystem of interventions. Each intervention has different consequences based on the timing and sequence of implementation. Moreover, many interdependencies exist between existing measures and new interventions. [43] One intervention is likely to have different outcomes in different healthcare systems.
..
Therefore, we recommend future policy effectivity studies to include a careful assessment of (external) validity of their results; an extensive description of the healthcare system, governmental regularization and market properties.”

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Reviewer: 2
Dr. T. Joseph Mattingly II, University of Maryland School of Pharmacy

Comments to the Author:
First, thank you for submitting this review to the journal. Consider the following to help advance your contribution through the peer review process:

Response: we thank dr. Mattingly II for his comments, which have certainly helped us to improve the quality of this manuscript.

1. Title/Objective - This article is described as an "international benchmark systematic review" which is very confusing since the methods include articles that merely compare drug prices. A more accurate reflection may simply be a review of drug price comparison studies. Especially since you included the "national" studies that don't really compare across countries (e.g. How does a phone survey of 13 pharmacies by Scott et al. asking for the "cash"/retail price meet your inclusion criteria?).

Response: we have removed the words ‘benchmark’ and ‘international’ from the manuscript and have changed the title as suggested by the reviewer. As we aimed to give a complete overview of drug price comparison studies, national setting of a study was not an exclusion criterion.
Although the method may seem slightly inornate, we believe that asking for cash prices reflects actually paid (net) prices (in this instance: directly by the consumer) and is therefore justly included.
To improve clarity, we now discuss the national studies separately from the international studies to increase clarity for readers.
In addition, the national studies are now clearly included separately at the end of table 1 (study overview).

2. Grey Literature - Peer-reviewed academic journal articles may not be the most appropriate source of information regarding drug pricing comparisons. Government, non-profit organization, and other organizational reports may actually be more valuable than evidence aimed at academics.

Response: we recognize that governments and NGO’s often have excellent data sources. However, our aim was to assess academic evidence, written by independent observers and published with peer-reviewed standards. Often, however, our most highly rated studies were collaborations between governments/NGO’s and research institutions. We now highlight the value of these data and these collaborations in our discussion section.

Manuscript changes (Methods):
“We did not include primary grey literature documents, such as documents of non-governmental organizations, as our aim was to assess academic evidence of modifiable determinants, written by independent observers and published in peer-reviewed journals.”

Manuscript changes (Discussion):
“ The lack of net pricing is likely partly due to the fact that we excluded primary grey literature documents from our main research corpus. Various institutions, most prominently the WHO, are increasing their efforts in assessing pharmaceutical expenditure and its determinants as shown by a recent overview of grey literature. [49]
Research institutes have set-up various inspiring collaborations with governmental and non-governmental organizations (NGO) using national databanks (such as Austria’s Pharma Price Information, or PPI) or World Health Organization (WHO) data, which are rich and often available at request. [32, 50, 51] In addition, WHO now promotes and collects various drug price databases and occasionally publishes in peer-reviewed journals. [32, 52] Ideally, these studies give rise to further

collaboration and NGO-supervised observational studies in developing countries, allowing academics to assess effectiveness of interventions on net drug prices."

3. Retail vs. Out-of-pocket Price - Scott et al. 2019 and Theisen et al. 2019 literally did the same thing: "called pharmacies to ask for price for a cash-paying patient". However, you categorize them differently? Why?

Response: we thank the reviewer for his attentiveness. We have chosen to use the price category term as used in the original article if possible and unambiguous and have added this as footnote to the table. However, we agree with the reviewer that for this case, this might be confusing. Therefore, we have changed the price category of Scott et al. to out-of-pocket price, as we believe this is a widely used and more specific term than 'retail price'. In addition, we have clarified the determination of a study's price category in the table's legend.

Manuscript changes (Table 1 legend):
"Price category listed as mentioned or as described in the article."

4. Study Quality - Scott et al. 2019 made phone calls to pharmacies across 5 US cities (Boston, Charlotte, Denver, Detroit, and Seattle) but made no price adjustments based on purchasing power. Based on your stated quality rating system, it seemed a bit surprising this meet all quality criteria to earn a "good" rating (same for Theisen et al).

Response: Indeed, one question regarding covariates and statistical adjustment was positively rated in both studies, as it concerns two national studies and these out-of-pocket prices are an example of the actually paid prices, which is rare in this research. However, we understand the reviewer's concerns, and have therefore sharpened our quality rating regarding price adjustment for both studies and, accordingly, the results section.

5. Conclusions - "Uniform introduction of pricing regulation and publication of net drug prices will lead to more price transparency..." - How do you conclude this based on what you did? You reviewed peer-reviewed papers of other people who compared drug prices. While the statement may be true, I'm confused how this statement made it into the conclusion of your systematic review. It sounds like something that is more appropriate in your discussion.

Response: Indeed, market or pricing transparency was not selectively measured in the included studies. Therefore, we have moved this section of the Conclusion to our Discussion as suggested by the reviewer with corresponding evidence of the effect of pricing transparency.

Manuscript changes:
Discussion:
"Uniform introduction of pricing regulation and publication of net drug prices may ultimately lead to more price transparency and a more balanced playing field of countries with a significant budget constrain. Such a situation allows for fair negotiations and ultimately a more sustainable and transparent healthcare sector."

Conclusion: "We call for openness of net drug price data, as this is our key to change the healthcare cost burden."

VERSION 2 – REVIEW

REVIEWER	Mattingly II, T. Joseph University of Maryland School of Pharmacy, Pharmacy Practice and Science
REVIEW RETURNED	02-Mar-2021

GENERAL COMMENTS	Thank you for thoroughly addressing all previous comments. This revision provides a good contribution to the literature on drug pricing.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2
Dr. T. Joseph Mattingly II, University of Maryland School of Pharmacy
Comments to the Author:
Thank you for thoroughly addressing all previous comments. This revision provides a good contribution to the literature on drug pricing.

Competing interests of Reviewer: None declared

Response:

We thank the reviewer for his kind words and guidance throughout the entire process.